

Chapter 1: Algebra

Algebra is a branch of mathematics dealing with symbols and the rules for manipulating those symbols. A linear equation is an equation of the first degree, like $2x + 5 = 13$.

A quadratic equation is a polynomial equation of degree 2. The standard form is $ax^2 + bx + c = 0$, where a is not 0. The quadratic formula is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Example: Solve $x^2 - 5x + 6 = 0$. Factoring gives $(x-2)(x-3) = 0$, so $x = 2$ or $x = 3$.

Chapter 2: Calculus

Calculus is the mathematical study of continuous change, divided into differential calculus and integral calculus. Differentiation finds the derivative, the rate of change. The Power Rule: $\frac{d}{dx}(x^n) = n \cdot x^{(n-1)}$.

Integration is the reverse of differentiation. The indefinite integral of $f(x)$ is denoted by $\int f(x) dx$. For x to the power n , the integral is $x^{(n+1)}/(n+1) + C$.

Chapter 3: Geometry

Geometry is the study of shapes, sizes, and properties of space. A circle is the set of points equidistant from a center. Area = πr^2 , Circumference = $2\pi r$.

The Pythagorean Theorem: in a right-angled triangle, $a^2 + b^2 = c^2$, where c is the hypotenuse.